

Daryl Ng

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SKILLS

Languages	Dart, Kotlin, Java, JavaScript, C, SystemVerilog, Assembly
Technologies	Google Maps, REST API, AWS, Flutter, SQL, MongoDB, Jest, React, Node.js, Express.js, Jetpack Compose
Other	MVC, MVVM, BLoC, Microsoft Azure, Agile, Git, Android SDK, Gradle, Postman, Jira, Unix, Linux

WORK EXPERIENCE

Software Engineer	Sep 2022 – Jun 2023
New Earth Ventures Inc	San Luis Obispo, CA

- Developed a cross-platform package delivery management application in iOS, Android, and Web for using Flutter, Jira, Git to meet OOP MVC design specs for 2 clients in an Agile environment

Embedded Google Maps SDK and API

- Proposed and integrated Google Maps with real-time GPS using JSON data from Future HTTP requests, enabling seamless in-app navigation for users and improving overall user experience
- Reduced API call costs by at least 50% by optimizing API connections between Google Maps APIs, the frontend, controller, and cloud database, improving both app performance and cost

Frontend AWS Login and Sign-Up

- Deployed cloud platforms AWS Cognito and AWS Amplify using Java to create, authenticate, and authorize user accounts, improving app security and automating user management
- Minimized backend requests when fetching data using a Singleton design pattern to cache non-sensitive account information, reducing server load and increasing efficiency
- Created and tested user interfaces with Bloc (Business Logic Components) and Bloc Test that are isolated, reusable, and tested for other developers to easily build from and implement

PROJECTS

Full-Stack Kotlin User Interface	Kotlin, Jetpack Compose, MVVM, Node.js, MongoDB, Express.js, Jest
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- Built a Full-stack user interface using Kotlin, Jetpack Compose, Material 3, Android SDK for the Frontend and Node.js, Express.js for the Backend with a MongoDB database to store user information
- Implemented RESTful APIs to make HTTP requests using Coroutine, MVVM, Jest, Test-driven development

Full-Stack React Pokémon Wiki	JavaScript, React, Node.js, Jest, MongoDB, Express.js, Tailwind, Microsoft Azure
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- Led a team of 4 developers to design and develop a web application for a Wikipedia that keeps track of Pokémons using JavaScript, React, Node.js, Tailwind, CSS, and Figma in an Agile environment
- Utilized Test-driven development of Backend RESTful API endpoints to increase coverage by 80% using Jest, MongoDB, Express.js
- Deployed CI/CD pipelines using GitHub Actions and Azure to automatically verify any build errors

Multiprocessing Remote UDP File Sharing	C, POSIX, Unix, UDP, BSD Sockets
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- Independently designed and developed servers and clients for a multi-user remote file sharing system using C, Unix/Linux, and UDP sockets, enabling simultaneous file transfers
- Implemented robust packet transmission protocols using Circular Queues with Sliding Window and Selective-Repeat ARQ algorithm, to increase efficiency, error detection, and recovery
- Drafted packet flow and state machine diagrams to simplify and organize the system and code

EDUCATION

California Polytechnic State University, San Luis Obispo	Jun 2023
Bachelor of Science in Computer Engineering	